

Chang-an Lu

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Research Interests

Computer Graphics; 3D Reconstruction and Generation; Structured Explicit/Implicit 3D Representations; Octree-based Neural Geometry; Physics-Grounded World Models; Programmatic and Neurosymbolic 3D Scene Generation.

Education

Peking University, School of Electronics Engineering and Computer Science Expected June 2027
B.Sc. in Information and Computing Science
Major GPA: 3.56/4.0; Overall GPA: 3.456/4.0

Research Experience

Neural Explicit Shape Representation

Sep. 2025 – Present

Advisor: Prof. Peng-Shuai Wang

Submitted to ICLR 2027

- Worked on explicit 3D shape representation and compression via ray-surface intersection, responsible for the single-object optimization module.
- Designed a cyclic sampling strategy that iteratively generates sampling points from predicted surface positions, making full use of voxel corner features and enabling high-precision reconstruction with a smaller parameter budget.
- Developed an interleaved ray inference scheme that reorganizes ray-voxel pairs, enlarges effective batch size, and significantly reduces redundant computation.

Selected Course Projects

- **Qt-based Terrain Generation and Interaction**
[Code](#)
- **OpenGL-based Arbitrary-dimensional Poisson Disk Sampling**
[Code](#)
- **LoTR-based LLM Fine-tuning: Incorporating Attention Head Dimensions into Tensor Decomposition**
[Code](#)
- **Tensorized Design of the Overcooked Environment**
[Code](#)

Selected Coursework

- Introduction to Computing: 93; Programming Practicum: 94; Data Structures and Algorithms: 88
- Introduction to Machine Learning: 91; Frontiers of Geometric Computing: 93.5; Advanced Rendering: 98
- Parallel Algorithm Design: 96.5; LLM: Principles, Challenges, and Industrial Applications: 98

- Large Language Models and Agents: A; Graph Neural Networks: A+

Awards and Competitions

- National Second Prize, Chinese Mathematical Olympiad (CMO), 2020
- Qin Wanshun Jin Yunhui Scholarship, Peking University, 2023
- Beijing Second Prize, National College Mathematical Modeling Competition, Sep. 2025
- Excellence Award, Medical Image Segmentation Challenge (PKU School of Physics $\times$ PKU Third Hospital), May 2025

Other Experience

- VCL Summer School, Jul. 2025 – Sep. 2025
- Teaching Assistant, Computer Vision, Peking University, Sep. 2025 – Jan. 2026

Skills

- Programming: C++, Python, CUDA
- Graphics/Geometry: libigl, Eigen, OpenGL, Blender
- Machine Learning: PyTorch
- Mathematics: Linear Algebra, Numerical Optimization
- Tools: Git, CMake, LaTeX, Linux